# Weekly Report: Remote Data Sharing and Memory-Hierarchy Bottlenecks on Wafer-Scale GPUs

Week ending June 1, 2026

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## Abstract

This report summarizes one week of work on understanding data sharing and remote memory behavior on wafer-scale GPUs. The main outcome is a shift from a broad "data-sharing-aware scheduling" idea to a more precise remote-access taxonomy. Across traditional and ML workloads, high remote-access ratio is not by itself evidence of a useful data-sharing optimization. Remote traffic can come from private placement mismatch, output/write ownership mismatch, shared read-mostly reuse, or topology-local neighbor reuse. New traces were added to classify these cases at page granularity and to attribute remote reads to remote L2/MSHR or remote DRAM service.

## Keywords

wafer-scale GPU, remote memory, data sharing, L2 cache, bottleneck analysis

## 1 Summary

This week's work focused on turning page-sharing traces into architectural evidence. The main question was whether remote data access is a true bottleneck, and if so, whether the right mechanism is page placement, home-L2 caching, or a neighbor-exposed L2 hierarchy.

The strongest result is that remote accesses are heterogeneous. For example, SpMV has high remote traffic, high remote-DRAM service, high neighbor locality, and a strong all-local oracle speedup. This makes it the best current case for remote data being on the critical path. PageRank is different: its remote traffic is more shared-read-mostly and mostly served by remote L2/MSHR, making it a cleaner home-L2 candidate. MatMul and AES are important negative controls: they have high remote and sharing ratios, but the prior all-local oracle did not improve performance.

## 2 Complete Work Log

### 2.1 Research Framing

The initial idea was to exploit cross-tenant data sharing in RAG-style multi-tenant serving by grouping thread blocks around shared read-mostly data. After looking at the available traces, the direction changed. The current benchmarks do not yet expose a clean tenant/request boundary, and the observed data reuse is easier to reason about at page and GPU level than at thread-block level. The working framing is now:

> Remote accesses on wafer-scale GPUs should first be classified. Only a subset of remote traffic is true shared read-mostly reuse; other traffic is placement or output ownership mismatch.

Related work was also re-positioned. STAR is useful as a multi-tenant benchmark-construction reference, but its main axis is destructive interference in partitioned GPU environments. This project instead targets constructive reuse and topology-local remote data movement. LADM is closer because it studies locality-centric data and threadblock management for massive GPUs, but the current traces do not yet provide an obvious thread-block level hook. This motivated the page-level and GPU-level trace work below.

### 2.2 Experiment and Visualization Work

The following analysis and visualization work was completed this week.

- Generated chord diagrams and row-style requester-owner diagrams for remote access flows, including versions that cover 80% of remote requests.
- Updated traditional wafer-flow maps so that all traditional workloads are plotted instead of a hand-selected subset.
- Added remote-access ratio annotations to the row-style diagrams.
- Added remote-page local-use analysis: whether a page remotely accessed by GPU A is also locally used by its owner GPU.
- Built an all-local data-access bottleneck copy of the codebase while keeping remote address translation through IOMMU. This is used as an oracle to ask whether remote data movement is on the performance path.
- Compared bottleneck-analysis results against the original result directory only after checking problem signatures.
- Added bar plots for speedup-over-baseline and remote-access ratio.
- Added the new L2-source page-service trace and generated the latest traditional result set with it.

### 2.3 Result Artifacts

## 3 Instrumentation Completed

### 3.1 Page-Level Sharing Trace

The page-sharing trace now records both virtual and physical page information: `pid`, `page_id`, `page_vaddr`, `page_paddr`, `page_block`, `page_size`, `paddr`, `bytes`, and `is_write`. Each run now emits a per-page file, `*_sharing_pages.csv`, which summarizes read/write ratio, local and remote accesses, sharer count, remote sharer count, dominant owner, dominant requester, requester counts, owner counts, and reuse window.

**Table 1: Main result directories and report artifacts produced or used this week.**

| Artifact | Path |
|---|---|
| Traditional sharing traces | `akkalat/results/2026-05-26-01-59-52-translation-sweep` |
| Traditional L2-source run | `akkalat/results/2026-06-01-00-17-43-sampled-validation` |
| LLM sampled validation | `akkalat/results/2026-05-29-03-05-55-sampled-validation` |
| Traditional all-local oracle | `bottleneck analysis/akkalat/results/2026-05-29-18-10-44-traditional-bottleneck-analysis` |
| LLM all-local oracle | `bottleneck analysis/akkalat/results/2026-05-31-04-22-59-traditional-bottleneck-analysis` |
| Standalone report | `akkalat/results/weekly_report_2026-06-01` |
| ACM-template report | `weeklyreport/06012026` |

**Table 2: Selected traditional workloads. Remote L2 includes remote L2 cache hits and remote MSHR service. Neighbor means 1-hop requester-provider traffic. Shared-read and private are fractions of remote accesses after page-level classification.**

| Benchmark | Remote | Remote L2 | Remote DRAM | Neighbor | Shared-read | Private |
|---|---|---|---|---|---|---|
| spmv | 66.3 | 18.9 | 81.1 | 65.5 | 6.8 | 64.4 |
| pagerank | 31.8 | 91.8 | 8.2 | 18.3 | 70.9 | 29.1 |
| fastwalshtransform | 70.5 | 96.1 | 3.9 | 60.1 | 0.0 | 52.8 |
| fir | 69.5 | 91.1 | 8.9 | 70.7 | 1.5 | 93.7 |
| matrixmultiplication | 89.0 | 100.0 | 0.0 | 5.2 | 97.6 | 0.0 |
| aes | 94.8 | 6.3 | 93.7 | 80.0 | 99.8 | 0.2 |
| kmeans | 71.7 | 93.6 | 6.4 | 16.7 | 0.0 | 95.2 |
| stencil2d | 97.4 | 34.6 | 65.4 | 7.4 | 63.5 | 0.0 |
| nw | 97.9 | 25.1 | 74.9 | 5.7 | 12.3 | 34.3 |

## 3.2 L2 Service-Source Trace

The L2-source tracer now emits a compact page-level service file: `*_l2_source_page_source.csv`. The main fields are: `source`, `requester_gpm`, `provider_gpm`, `hops`, `is_neighbor`, `access_type`, `page_paddr`, `accesses`, `bytes`, `avg_latency_ns`, `first_time_ns`, and `last_time_ns`. This makes it possible to classify remote reads as remote L2 cache hits, remote MSHR service, remote DRAM service, or 1-hop neighbor service.

## 3.3 Analysis Scripts

The new post-processing script `analyze_l2_home_evidence.py` joins page-sharing summaries with L2-source traces and emits: `l2_home_evidence_pages.csv` and `l2_home_evidence_summary.csv`. The key output metrics are remote-read L2 service ratio, remote-read DRAM service ratio, neighbor-read ratio, and a page-level classification into shared-read-mostly, owner-and-remote use, remote-private mismatch, write/output mismatch, and local-only pages.

# 4 Traditional Workload Results

Table 2 and Figures 1–3 summarize the newest traditional run. The result directory is `akkalat/results/2026-06-01-00-17-43-sampled-validation`.

## 4.1 Key Traditional Cases

**SpMV** is the strongest bottleneck candidate. It has 66.3% remote accesses, 81.1% remote reads served by remote DRAM, 65.5% neighbor remote reads, and a prior all-local oracle speedup of 3.58×. This suggests that remote data movement is on the critical path, but the traffic is not purely shared read-mostly; much of it looks like placement or output mismatch.

**PageRank** is the cleanest shared-read/home-L2 candidate. It has 31.8% remote accesses, 70.9% shared-read-mostly remote traffic, 91.8% remote L2/MSHR service, and a prior all-local oracle speedup of 2.38×.

**FWT** and **FIR** are good neighbor-L2 candidates. They have high remote ratios, most remote reads are served by remote L2/MSHR, and neighbor remote reads are 60.1% and 70.7%, respectively. These results motivate an L2 design that isolates local bandwidth from neighbor-facing bandwidth.

**MatMul** and **AES** are useful negative controls. MatMul has 89.0% remote accesses and 97.6% shared-read-mostly remote traffic, but remote reads are almost entirely served by L2/MSHR and prior all-local speedup is about 1.0×. AES also has high remote and shared-read ratios, but prior all-local speedup is about 1.0×. These cases show that high remote ratio and high sharing do not automatically imply performance opportunity.

## 4.2 All Traditional Results

Table 3 reports the full traditional L2-source classification result. The columns are: total remote-access ratio; remote-read service from L2/MSHR; remote-read service from DRAM; 1-hop neighbor remote-read ratio; and four page-level classes as fractions of remote accesses.

## 4.3 Traditional Bottleneck Oracle

Table 4 summarizes the comparable all-local oracle runs from the bottleneck-analysis copy. The "Speedup" column is old baseline time divided by all-local oracle time. A value near 1.0 means that removing remote data movement did not improve runtime under this experiment.

The main takeaways from the oracle are:

- **Remote data is performance-critical for SpMV, PageRank, KMeans, FWT, and FIR.** SpMV is the most important case because its speedup is 3.58× and most of its remote reads go to remote DRAM.
- **Remote access is not automatically performance-critical.** Bitonic, FloydWarshall, SimpleConv, AES, and MatMul have speedups close to 1.0× even when their remote-access ratios can be high.
- **The optimization target differs by workload.** SpMV points to data placement or neighbor service. PageRank

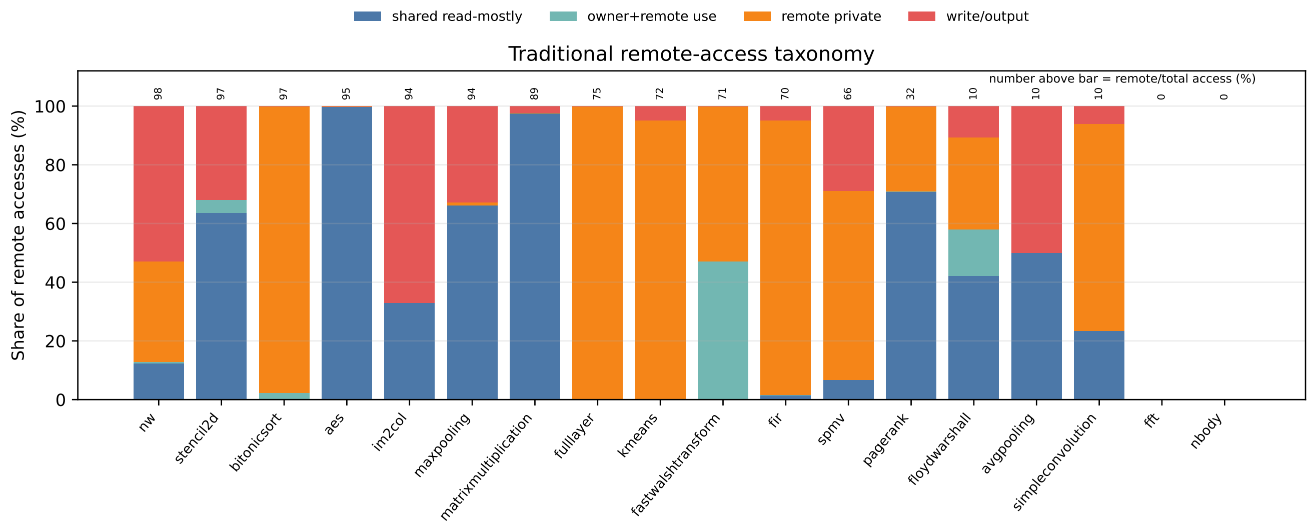


**Figure 1: Remote-access taxonomy for traditional workloads. The number above each bar is the total remote-access ratio. High remote ratio can mean very different things: shared read reuse, private placement mismatch, output/write mismatch, or a mixture.**

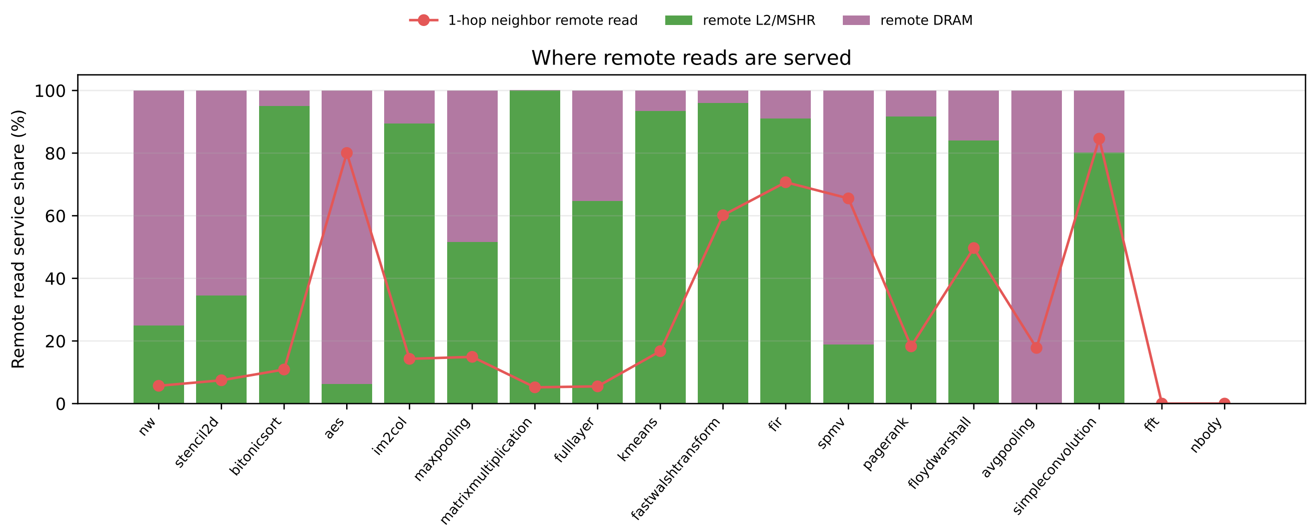


**Figure 2: Remote read service source. Some workloads are mostly served by remote L2/MSHR, while others still send a large fraction of remote reads to remote DRAM. The red line shows the share of remote reads that are 1-hop neighbor accesses.**

points to shared-read home-L2. FWT and FIR point to topology-local neighbor-facing L2 service. MatMul and AES should be used as negative controls.

## 5 LLM / ML Workload Results

The LLM operators show high remote ratios and, for matrix-heavy operators, high shared-access ratios. Attention Q/K/V/out and MLP FC1/FC2 are the most important operators to inspect next because they combine long runtime with high remote traffic. However, the current LLM all-local comparison produced 1.0× speedup for the comparable completed operators. This means the current evidence is not sufficient to claim that LLM remote data access is on the critical path.

The next step for ML workloads is therefore not just more remote-ratio plots. We need stall attribution and tensor semantics: whether a remote page is a weight, activation input, activation output, temporary, embedding, or KV/cache page. Without tensor labels, page-level sharing can mix weight

**Table 3: Full traditional workload remote-access taxonomy from the latest L2-source run.**

| Benchmark | Remote | L2 | DRAM | Neighbor | Shared | Owner+R | Private | Write |
|---|---|---|---|---|---|---|---|---|
| nw | 97.9 | 25.1 | 74.9 | 5.7 | 12.3 | 0.6 | 34.3 | 52.9 |
| stencil2d | 97.4 | 34.6 | 65.4 | 7.4 | 63.5 | 4.6 | 0.0 | 31.9 |
| bitonicsort | 97.2 | 95.2 | 4.8 | 10.9 | 0.0 | 2.4 | 97.6 | 0.0 |
| aes | 94.8 | 6.3 | 93.7 | 80.0 | 99.8 | 0.0 | 0.2 | 0.0 |
| im2col | 94.3 | 89.6 | 10.4 | 14.3 | 33.0 | 0.0 | 0.0 | 67.0 |
| maxpooling | 94.2 | 51.6 | 48.4 | 14.9 | 66.2 | 0.0 | 1.0 | 32.9 |
| matrixmultiplication | 89.0 | 100.0 | 0.0 | 5.2 | 97.6 | 0.0 | 0.0 | 2.4 |
| fulllayer | 75.2 | 64.8 | 35.2 | 5.5 | 0.0 | 0.0 | 100.0 | 0.0 |
| kmeans | 71.7 | 93.6 | 6.4 | 16.7 | 0.0 | 0.0 | 95.2 | 4.8 |
| fastwalshtransform | 70.5 | 96.1 | 3.9 | 60.1 | 0.0 | 47.2 | 52.8 | 0.0 |
| fir | 69.5 | 91.1 | 8.9 | 70.7 | 1.5 | 0.0 | 93.7 | 4.8 |
| spmv | 66.3 | 18.9 | 81.1 | 65.5 | 6.8 | 0.0 | 64.4 | 28.8 |
| pagerank | 31.8 | 91.8 | 8.2 | 18.3 | 70.9 | 0.0 | 29.1 | 0.0 |
| floydwarshall | 10.3 | 84.0 | 16.0 | 49.7 | 42.2 | 15.9 | 31.2 | 10.7 |
| avgpooling | 10.2 | 0.0 | 100.0 | 17.8 | 50.0 | 0.0 | 0.0 | 50.0 |
| simpleconvolution | 9.5 | 80.3 | 19.7 | 84.6 | 23.5 | 0.0 | 70.5 | 6.0 |
| fft | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| nbody | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |

**Table 4: Traditional all-local data oracle results. Translation still goes through the IOMMU, so this isolates data movement more than address translation.**

| Benchmark | Speedup | Remote | Dist. | Top3 dir. | Private remote | Owner-local remote |
|---|---|---|---|---|---|---|
| spmv | 3.58 | 66.3 | 1.42 | 92.2 | 99.5 | 0.5 |
| pagerank | 2.38 | 31.7 | 1.21 | 33.1 | 28.6 | 71.4 |
| kmeans | 1.99 | 71.7 | 2.55 | 17.6 | 100.0 | 0.0 |
| fastwalshtransform | 1.29 | 70.5 | 1.58 | 80.3 | 52.8 | 47.2 |
| fir | 1.15 | 69.5 | 1.42 | 96.4 | 99.1 | 0.9 |
| floydwarshall | 1.01 | 10.3 | 0.34 | 64.4 | 41.9 | 58.1 |
| bitonicsort | 1.01 | 97.2 | 3.60 | 21.9 | 97.6 | 2.4 |
| simpleconvolution | 1.00 | 9.5 | 0.17 | 100.0 | 76.5 | 23.5 |
| matrixmultiplication | 1.00 | 89.0 | 5.32 | 6.3 | 2.4 | 97.6 |
| aes | 1.00 | 94.8 | 5.79 | 6.4 | 0.2 | 99.8 |

broadcast, output placement, temporary buffers, and true shared read-mostly reuse.

## 5.1 All LLM Operator Results

Table 6 shows all 16 decomposed GPT-7B operators from the sampled-validation run used for the current LLM analysis. The matrix-heavy operators have high remote access and often high shared-access ratios, but this alone is not a bottleneck proof.

## 5.2 LLM Bottleneck Oracle

Table 7 reports the completed comparable LLM all-local-oracle comparison. The speedups are all 1.0× in this run. This does not mean remote data can be ignored. It means

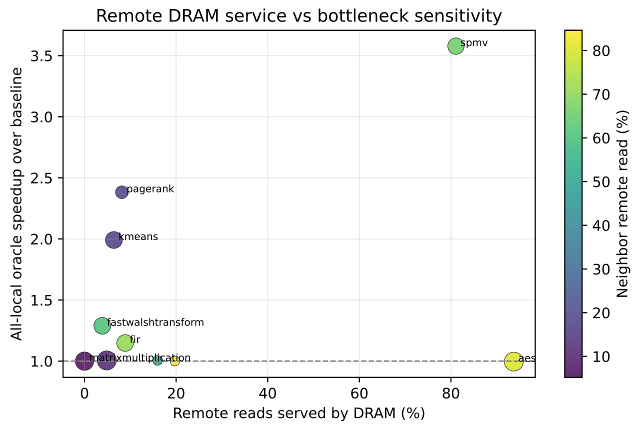


**Figure 3: Remote DRAM service compared with prior all-local oracle speedup. SpMV is the strongest remote-data bottleneck. PageRank also benefits, but its remote reads are mostly served by remote L2/MSHR, suggesting a different mechanism.**

**Table 5: Representative LLM decomposed operators from the GPT-7B sampled validation run.**

| Operator | Time (us) | Remote | Shared access | Shared page | L1 miss | L2 miss |
|---|---|---|---|---|---|---|
| embedding | 9.5 | 95.5 | 7.7 | 0.1 | 100.0 | 99.8 |
| layer00_attn_q | 10941.0 | 94.0 | 99.5 | 88.2 | 75.6 | 21.8 |
| layer00_attn_score | 1919.5 | 96.7 | 97.4 | 92.1 | 98.0 | 1.8 |
| layer00_attn_value | 164.0 | 95.1 | 99.8 | 90.7 | 26.7 | 33.3 |
| layer00_mlp_fc1 | 25223.0 | 92.7 | 99.0 | 81.1 | 71.7 | 28.5 |
| layer00_mlp_fc2 | 6842.2 | 94.6 | 38.0 | 45.0 | 30.3 | 17.9 |
| layer00_norm1 | 38.5 | 98.0 | 0.0 | 0.0 | 50.0 | 99.3 |
| layer00_attn_residual | 8.6 | 94.1 | 0.0 | 0.0 | 100.0 | 100.0 |

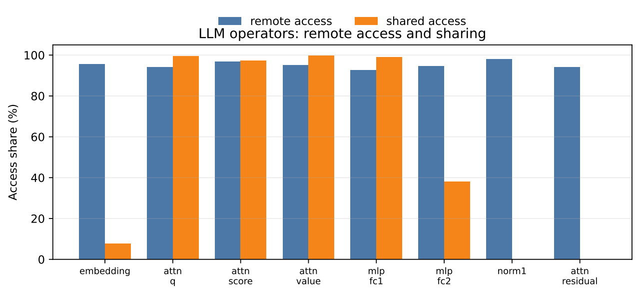


**Figure 4: LLM operators often have both high remote-access ratio and high shared-access ratio, especially attention projections and MLP fully-connected operators.**

the current oracle does not expose a performance sensitivity for these operators. The next LLM run therefore needs direct stall attribution and page-to-tensor semantic labels.

### 5.3 Small LLM Home-L2 Evidence

A small GPT-7B embedding smoke test produced one concrete shared-read page with 32 requester GPUs, 2048 total read accesses, 1984 remote accesses, and 99.9% of remote reads served by remote L2/MSHR rather than remote DRAM. This is a good example of the target behavior for selective home-L2 caching. However, it is still a smoke-test result. The full LLM run should be repeated with the new page-source trace before this becomes a central result.

## 6 Research Direction

### 6.1 Remote-Access Taxonomy

The central direction is to first classify remote traffic and then apply the right mechanism:

- **Remote-private placement mismatch**: use compute-data co-placement or better initial page placement.
- **Remote write/output mismatch**: use output ownership or producer-consumer placement.
- **Shared read-mostly reuse**: use selective home-L2 caching.
- **Topology-local neighbor reuse**: use neighbor-exposed L2 service.

### 6.2 Selective Home-L2 Caching

Home-L2 caching should target pages with high read ratio, multiple sharers, enough remote reads, and low write activity. This avoids treating placement mismatch as data sharing.

### 6.3 Neighbor-Exposed L2

The more architecture-oriented idea is to split L2 service into a local path and a neighbor-facing path. The local path serves the owning GPU with high priority, while the neighbor path exposes selected lines/pages to north, south, east, and west neighbors through separate queues or bandwidth. This is not page migration and not global coherence. It is a bounded, topology-aware memory-hierarchy extension. SpMV, FWT, FIR, and SimpleConv provide the strongest early evidence for this path.

### 6.4 Cross-GPU Translation Prefetching

Cross-GPU translation prefetching was considered as an auxiliary mechanism. It can pre-populate translation and owner metadata for remote pages, reducing TLB and IOMMU lookup overhead. It does not by itself solve remote data movement or remote DRAM pressure. The current conclusion is that translation prefetching is best treated as support for home-L2 or neighbor-L2 policies, not as the main contribution.

### 6.5 Why MatMul Is a Negative Control

MatMul is useful precisely because it looks promising under a simple remote ratio metric but does not benefit under the all-local oracle. It has 89.0% remote accesses and 97.6% shared-read-mostly remote traffic, but remote reads are almost entirely served by L2/MSHR, the neighbor-read ratio is only 5.2%, and the oracle speedup is about 1.0×. This argues that future designs must be selective and performance-aware.

**Table 6: All decomposed GPT-7B operators in the current LLM analysis. Time is driver total time in microseconds.**

| Operator | Time | GPUs | CPI | L1 miss | L2 miss | Remote | Shared acc. | Shared pg. |
|---|---|---|---|---|---|---|---|---|
| embedding | 9.5 | 43 | 14.0 | 100.0 | 99.8 | 95.5 | 7.7 | 0.1 |
| layer00_norm1 | 38.5 | 4 | 10.9 | 50.0 | 99.3 | 98.0 | 0.0 | 0.0 |
| layer00_attn_q | 10941.0 | 47 | 604.6 | 75.6 | 21.8 | 94.0 | 99.5 | 88.2 |
| layer00_attn_k | 10941.0 | 47 | 604.6 | 75.6 | 21.8 | 94.0 | 99.5 | 88.2 |
| layer00_attn_v | 10941.0 | 47 | 604.6 | 75.6 | 21.8 | 94.0 | 99.5 | 88.2 |
| layer00_attn_score | 1919.5 | 16 | 32.7 | 98.0 | 1.8 | 96.7 | 97.4 | 92.1 |
| layer00_causal_mask | 4.0 | 2 | 9.6 | 0.0 | 100.0 | 27.5 | 0.0 | 0.0 |
| layer00_attn_softmax | 7.9 | 4 | 5.4 | 33.3 | 99.4 | 75.0 | 0.0 | 0.0 |
| layer00_attn_value | 164.0 | 47 | 5.2 | 26.7 | 33.3 | 95.1 | 99.8 | 90.7 |
| layer00_attn_out | 10941.0 | 47 | 604.6 | 75.6 | 21.8 | 94.0 | 99.5 | 88.2 |
| layer00_attn_residual | 8.6 | 43 | 37.2 | 100.0 | 100.0 | 94.1 | 0.0 | 0.0 |
| layer00_norm2 | 38.5 | 4 | 10.9 | 50.0 | 99.3 | 98.0 | 0.0 | 0.0 |
| layer00_mlp_fc1 | 25223.0 | 48 | 1174.5 | 71.7 | 28.5 | 92.7 | 99.0 | 81.1 |
| layer00_mlp_gelu | 11.1 | 46 | 13.3 | 100.0 | 99.8 | 84.7 | 0.0 | 0.0 |
| layer00_mlp_fc2 | 6842.2 | 47 | 307.6 | 30.3 | 17.9 | 94.6 | 38.0 | 45.0 |
| layer00_mlp_residual | 8.6 | 43 | 37.2 | 100.0 | 100.0 | 94.1 | 0.0 | 0.0 |

## 7 Current Claims

Based on this week's evidence, the strongest claims are:

(1) Remote traffic on wafer-scale GPUs is heterogeneous; remote-access ratio alone is not a reliable optimization signal.
(2) SpMV currently provides the clearest evidence that remote data movement can be a performance bottleneck.
(3) PageRank provides the clearest evidence for shared-read home-L2 style reuse.
(4) FWT and FIR provide evidence for topology-local neighbor-facing L2 service.
(5) MatMul and AES are important negative controls, showing that high remote traffic or high sharing can still produce little performance sensitivity.
(6) For LLM operators, high remote and shared-access ratios are visible, but the current oracle does not prove remote data is on the critical path.

## 8 Next Week Plan

(1) Rerun the full LLM decomposed workload with `--trace-sharing` and `--report-l2-source`, then analyze `l2_home_evidence_summary.csv`.
(2) Add stall attribution: remote memory stall cycles, translation stall cycles, NoC/RDMA wait, and DRAM queue wait.
(3) Add tensor-semantic labels for ML pages: weight, input activation, output activation, temporary, embedding, and KV/cache.
(4) Implement a neighbor-L2 oracle in which 1-hop remote L2/MSHR service receives reduced latency or isolated bandwidth.
(5) Use SpMV, PageRank, FWT, FIR, MatMul, and AES as the core benchmark set: SpMV and PageRank as positive cases, FWT/FIR as neighbor cases, and MatMul/AES as negative controls.

## 9 Commands

### 9.1 Traditional Workload Trace

```
cd /home/daoxuanxu/dataSharing/WSG-baseline-model/akkalat

python3 runall2.py \
  --benchmarks traditional \
  --configs baseline \
  --disable-servers \
  --trace-sharing \
  --trace-sharing-sample 1 \
  --trace-sharing-max-records 0 \
  --report-l2-source \
  --l2-source-tile-width 7 \
  --max-workers 10
```

### 9.2 LLM Decomposed Trace

```
cd /home/daoxuanxu/dataSharing/WSG-baseline-model/akkalat

python3 runllm_decomposed.py \
  --model gpt \
  --profile gpt-7b \
```

**Table 7: LLM all-local data oracle comparison. The current comparable run did not show speedup, despite high remote and shared-access ratios.**

| Operator | Speedup | Remote | Shared acc. | Shared pg. | Accesses |
|---|---|---|---|---|---|
| embedding | 1.00 | 95.5 | 7.7 | 0.1 | 106496 |
| layer00_norm1 | 1.00 | 98.0 | 0.0 | 0.0 | 98304 |
| layer00_attn_q | 1.00 | 94.0 | 99.5 | 88.2 | 1000000 |
| layer00_attn_k | 1.00 | 94.0 | 99.5 | 88.2 | 1000000 |
| layer00_attn_v | 1.00 | 94.0 | 99.5 | 88.2 | 1000000 |
| layer00_attn_score | 1.00 | 96.7 | 97.4 | 92.1 | 1000000 |
| layer00_causal_mask | 1.00 | 27.5 | 0.0 | 0.0 | 568 |
| layer00_attn_softmax | 1.00 | 75.0 | 0.0 | 0.0 | 4096 |
| layer00_attn_value | 1.00 | 95.1 | 99.8 | 90.7 | 1000000 |
| layer00_attn_out | 1.00 | 94.0 | 99.5 | 88.2 | 1000000 |
| layer00_attn_residual | 1.00 | 94.1 | 0.0 | 0.0 | 98304 |
| layer00_norm2 | 1.00 | 98.0 | 0.0 | 0.0 | 98304 |
| layer00_mlp_fc1 | 1.00 | 92.7 | 99.0 | 81.1 | 1000000 |
| layer00_mlp_gelu | 1.00 | 84.7 | 0.0 | 0.0 | 176128 |
| layer00_mlp_fc2 | 1.00 | 94.6 | 38.0 | 45.0 | 1000000 |
| layer00_mlp_residual | 1.00 | 94.1 | 0.0 | 0.0 | 98304 |

```
--layers 1 \
--seq-len 128 \
--split-k 8 \
--configs llm_mixed \
--sampled-warmup 128 \
--sampled-granularity 512 \
--disable-servers \
--trace-sharing \
--trace-sharing-sample 1 \
--trace-sharing-max-records 1000000 \
--report-l2-source \
--l2-source-tile-width 7 \
--max-workers 10 \
--summarize
```